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Everything Over ?



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Reverse Engineering



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WHAT IS REVERSE ENGINEERING?

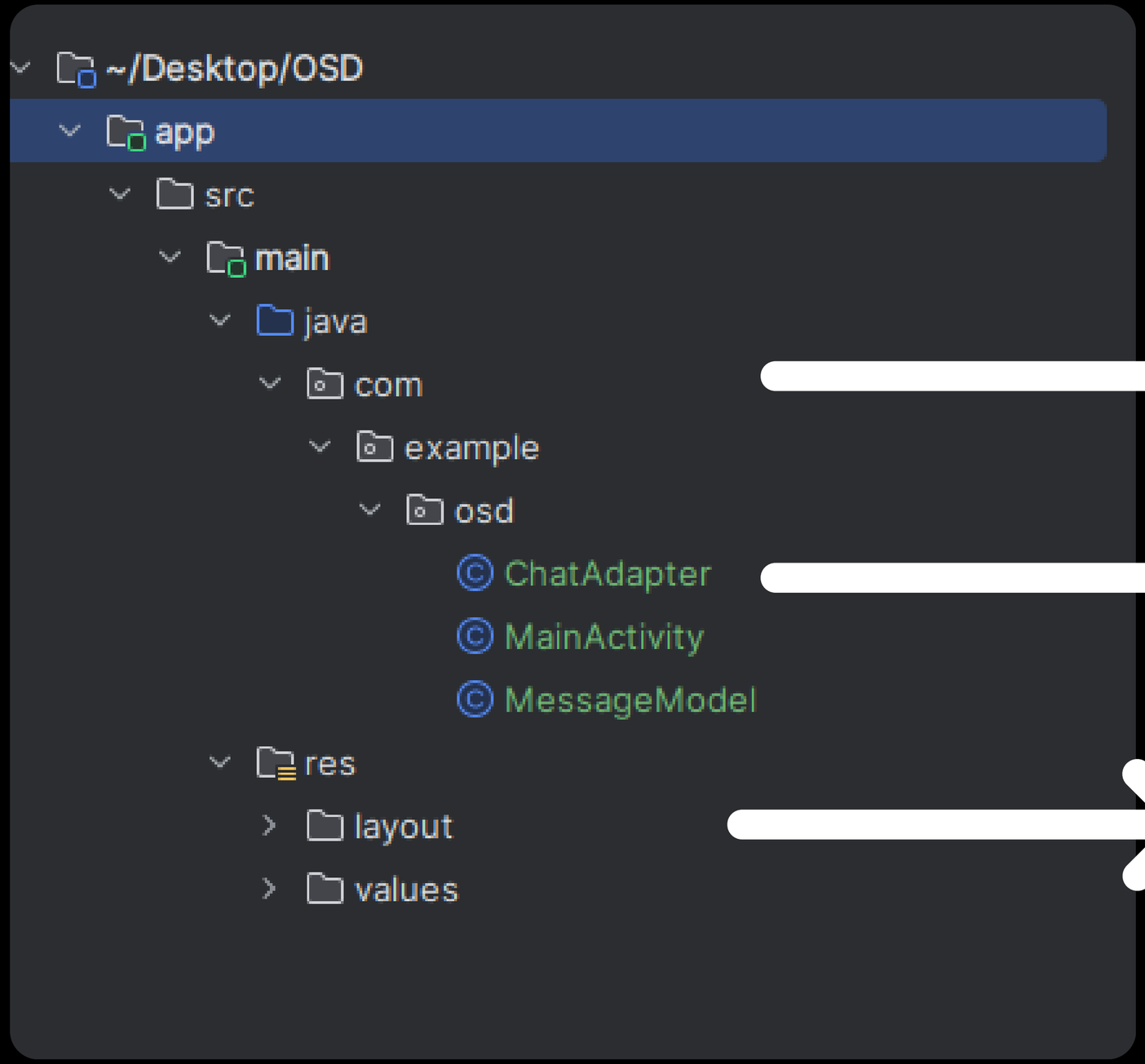
- Learning how something works by dismantling the **finished software** instead of building it yourself.



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Android Directory structure





Java Package

.java Files

.xml Layout Files

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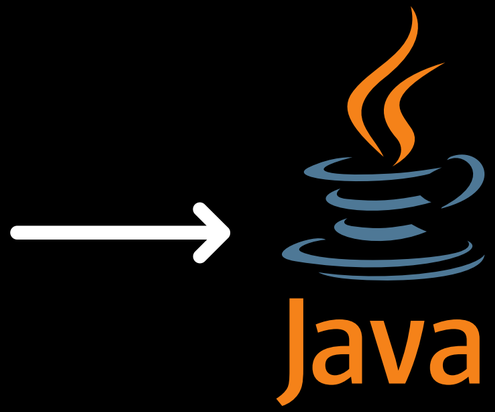
SOURCE CODE TO APK JOURNEY



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Java Source Code



Java Compiler



classes.dex



.apk File



.class Compiled File



Dex Compiler

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JADX

- JADX is an Android decompiler.
- It converts APK / DEX files into human-readable Java code.
- Reconstructed Java code is identical to the original code , it is not the exact copy.

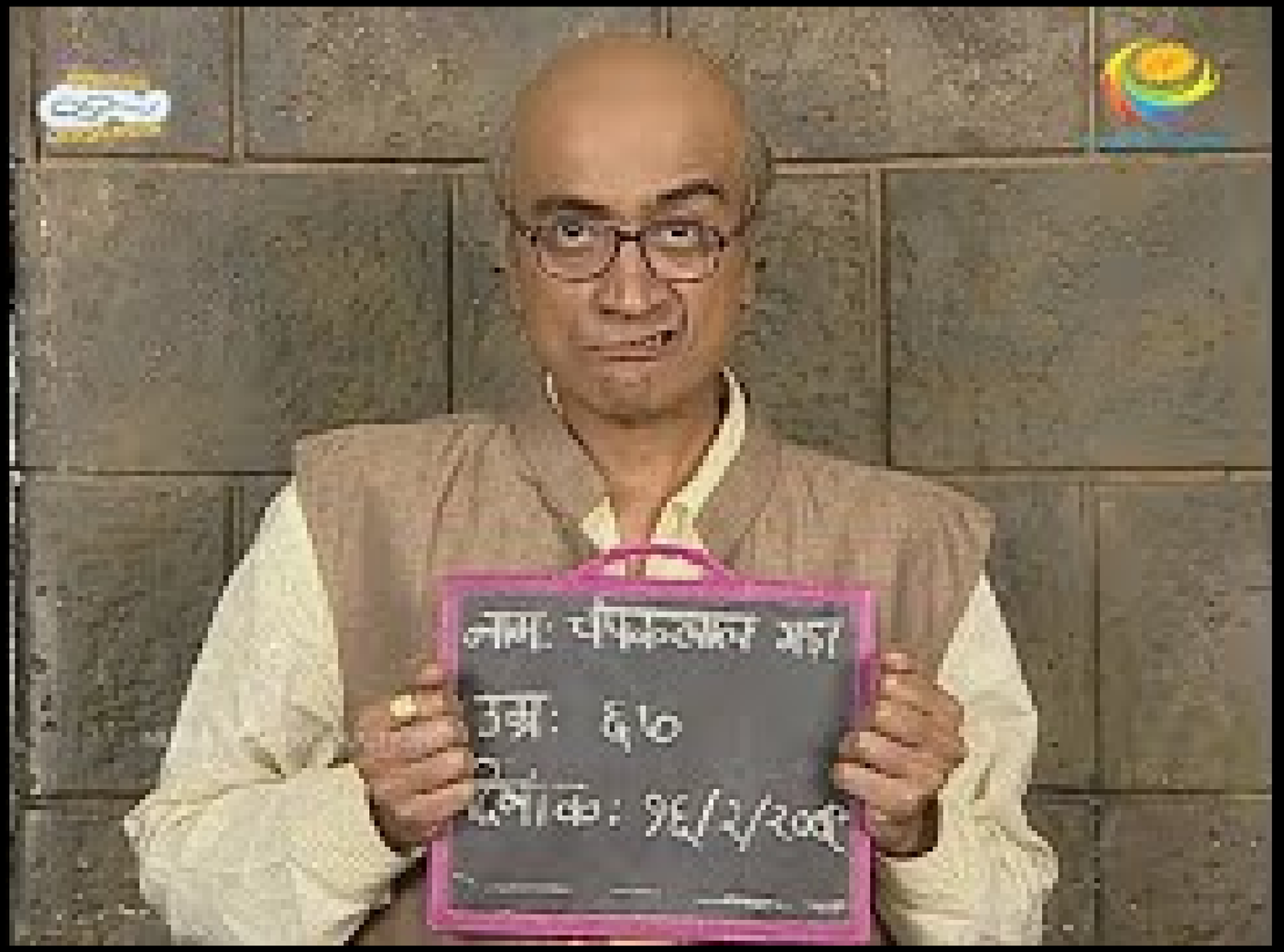


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⚠ Ethical & Legal Warning

- This tool is demonstrated only on **self-developed** Application for **Educational Purpose** Only !!
- **Reverse Engineering of Certified, Proprietary, or Government Applications without Explicit Authorization may violate Legal and Ethical Guidelines and is Strictly Discouraged.**

Or Else !!!



Command

```
jadx -d out osd2k26.apk
```

- **-d :** Flag used to set the Destination.
- **osd2k26.apk :** The APK to be Reverse Engineered.
- **out :** Destination Directory.

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Internal Working Of Jadx



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.apk



classes.dex



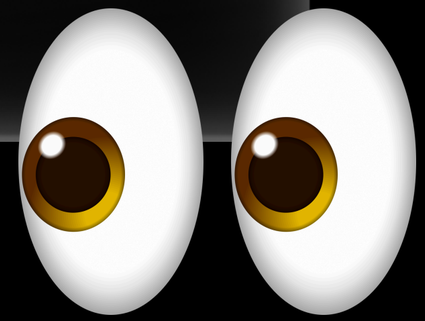
JADX



Reconstructed Java Code

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POV: Stealing Api's using Reverse Engineering



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EXPECTATIONS VS REALITY





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Protective measures



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Use a Backend as a Proxy :

Android App → Backend Server → Third-party API
The real API key lives ONLY on **Backend Server.**

Token-Based Authentication :

Instead of embedding keys, App logs in with username/password,
Backend issues a JWT/session token All future requests use that token.